• the means to enable the bridge team to handle alert announcements, and • the means to manage all alert related stat es in a distributed system structure in consistent manner.

18.2 If practicable, there should be not more than one alert fo r one situation that req uires attention.

18.3 The alert management should as a minimum be able to handle a ll alerts required by performance standards adopted by the Organization for navigational equipment comprised b y the INS or connected to the INS.

The alert management should have the capability to handle all other alerts of navigation al equipment comprised by the INS or connected to the INS in identical manner and should incorporate all alerts that are critical to the safety of navigation.

3 MSC.128(75), MSC.191(79).

RESOLUTION MSC.252(83) (adopted on 8 October 2007) ADOPTION OF THE REVISED PERFORMANCE S TANDARDS FOR INTEGRATED NAVIGATION SYSTEMS (INS)MSC 83/28/Add.3 ANNEX 30 Page 24 I:\MSC\ 83\28-Add-3.doc 18.4 The logical architecture of the alert mana gement and the handling concept for alerts - ADOPTION OF THE REVISED PERFORMANCE STANDARDS FOR INTEGRATED NAVIGA TION SYSTEMS (INS)MSC 83/28/Add.3 ANNEX 30 Page 24 I:\MSC\83\28-Add-3.doc 18.4 The logic al architecture of the alert mana gement and the handling concept for alerts should prov ide the capability to minimize the number of alerts especially those on a high priority level (e.g.

using system knowledge from redundancy concepts inside INS and evaluating inherent neces sities for alerts against navigational situat ions, operational modes or activated navig ational functions).

18.5 It should be possible to provide the centr al alert management HMI at least on the navigating and manoeuvring workstation and allowing the handling by the bridge team.

18.6 The audible announcement of alerts should en hance the guidance of the bridge team to the task stations or displays which are directly assigned to the function generating the alert and presenting the cause of the announcement and related information for decis ion support, e.g., dangerous target alarms should appear and have to be acknowledged at the workstation where the collision avoidance function is provided.

- presenting the cause of the announcement and related information for decision support, e.g., dangerous target alarms should appear and have to be acknowledged at the workstati on where the collision avoidance function is provided.

18.7 As alerts can be displayed at several locations , the system should be consistent a s far as practicable with respect to how alerts are displayed, silen ced and acknowledge d at any one task station of the INS.

19 Priorities and categories 19.1 Priorities of alerts 19.1.1 The alert management shoul d distinguis h between the three priorities listed: • Alarms • Warnings and • Cautions 1 9.1.2 Alarms should indicate conditions requiri ng immediate attention and action by the bridge team.

19.1.3 Warnings should indicate changed conditions and should be presented for precautio nary reasons wh